

1(a)	ref. to hazard and risk and precaution ;	<table><tr><th>hazard</th><th>risk</th><th>precaution</th></tr><tr><td>knife / blade / scalpel / AW</td><td>cuts / injury (skin / hand / AW)</td><td>cut away from the hand or cut onto a, board / tile / AW</td></tr><tr><td>avocado / catalase / enzyme / mesocarp</td><td>allergy / irritant</td><td>(wear) gloves / PPE</td></tr><tr><td>hydrogen peroxide / H₂O₂</td><td>irritant / toxic / corrosive / caustic / damage to eyes</td><td>(wear) gloves / goggles / safety glasses / PPE</td></tr></table>	hazard	risk	precaution	knife / blade / scalpel / AW	cuts / injury (skin / hand / AW)	cut away from the hand or cut onto a, board / tile / AW	avocado / catalase / enzyme / mesocarp	allergy / irritant	(wear) gloves / PPE	hydrogen peroxide / H ₂ O ₂	irritant / toxic / corrosive / caustic / damage to eyes	(wear) gloves / goggles / safety glasses / PPE	1 mark as prose ignore any other hazard ignore low / medium / high, risk R no risk allow cut away from body ignore cuts clothing / AW ; ignore gloves as precaution for knife ignore glasses / goggles ignore biohazard ignore hydrogen peroxide harmful ignore damage / (chemical) burns
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1(b)(i)	time / days, after picking / after harvesting / to ripen or level / degree / stages / AW, of ripeness ;		1 ignore ripeness / time, unqualified allow different as AW for level ignore age ; A colour of skin A softness of avocado / mesocarp ignore softness of skin												

Question	Answer	Marks	Guidance
1(b)(ii)	1 use of five different (stages)/AW of ripeness ; any six from: 2 same / stated / fixed, dimensions of avocado pieces ; 3 use knife / blade / scalpel / AW <u>and</u> ruler / AW (to cut the avocado pieces) or use knife / blade / AW <u>and</u> cork-borer <u>and</u> ruler (to cut cylinders) ;	7	ignore <i>ref. to</i> control experiment ignore 'known' for 'same' then ecf if repeated throughout allow standardised / controlled / fixed / constant A mesocarp as AW for avocado 1 A method to obtain 5 stages of ripeness allow if 5 different levels of ripeness stated even if less than 5 descriptions given – allow the mark ignore age 2 ignore removal of skin ignore size / mass / volume / SA for dimensions A cube with same / stated, dimensions / linear measurement(s) ignore length and width being the same unless 'cube' is mentioned cube 2 cm × 2 cm – BOD ignore any stated dimensions larger than 10 cm A cylinders same length and diameter (if numbers stated need to see units at least once) 3 allow cutter as AW for knife allow vernier calipers as AW for ruler

Question	Answer	Marks	Guidance
1(b)(ii)	4 use same, species / type / variety, of avocado or use same number / amount of pieces (in each beaker) ; 5 use a suitable method to maintain a constant temperature of hydrogen peroxide ; 6 use same / stated, volume / concentration of hydrogen peroxide / H_2O_2 ; 7 avocado added / AW to hydrogen peroxide and immediately start timer ; 8 suitable method / apparatus for measuring the quantity of oxygen produced / released ;		4 A <i>Persea americana</i> as same species A same tree A all pieces cut from same avocado A one/ a / the piece 5 suitable methods: water-bath temperature-controlled room A air conditioning / heater, set at one temperature ignore incubator / air conditioning unqualified if stated, temperature between 10 °C to 40 °C ignore concentration or volume values allow 'and then' as AW for immediately 8 A gas as AW for oxygen ignore count bubbles unqualified allow: counting bubbles using camera add washing up liquid and measure height of foam produced collect oxygen over water / H_2O_2 and measure volume in inverted measuring cylinder

Question	Answer	Marks	Guidance
1(b)(ii)	9 at each stage / AW of ripeness, set / measure / record / fixed period of, time (linked to release of oxygen) ; 10 at each stage of ripeness take at least three measurements ; 11 fresh / new / change / AW, hydrogen peroxide used in each experiment / different stages/AW of ripeness ;		collect oxygen and measure volume in a gas syringe ignore syringe allow measuring loss of oxygen by loss in mass, using an electric balance and mass change calculated allow idea of recording time to sink 10 ignore calculate a mean A three replicates / repeat 2 times, at each stage / AW of ripeness
1(c)(i)	any suitable method of generating random numbers and matching to the trees ;	1	e.g. number all the trees and use a random number generator (to select the trees) ignore quadrats / co-ordinates
1(c)(ii)	<i>idea of</i> to reduce the water loss from the avocados (while in storage) / AW ;	1	A reduces transpiration A reducing water evaporation allow no / stop / prevent AW for reduces allow losing moisture ignore 'keep moist' ignore vapour ignore prevent drying out ignore ref. to osmosis
1(c)(iii)	822 ± 412 and 736 ± 358 ;	1	

Question	Answer	Marks	Guidance						
1(c)(iv)	correct plotting of both bars (within ½ square) ;; 1 mark per bar	2	ecf for incorrect calculations from 1(c)(iii) if lines are thick or slanted take reading from top edge of bar <table><tr><td>group</td><td>expected range</td></tr><tr><td>20dR</td><td>410–1234</td></tr><tr><td>30dR</td><td>378–1094</td></tr></table>	group	expected range	20dR	410–1234	30dR	378–1094
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1(c)(v)	any four from: <i>mp2 only valid if mp1 has been awarded</i> <i>mp4 only valid if mp3 has been awarded</i> 1 <i>conclusion 0d compared with 10 – 30d or between 10 – 30d</i> effect of (cold storage only on catalase activity) is, variable / increases and decreases / fluctuates / no pattern / trend ; 2 <i>data comparison with 0d and 10 / 20 / 30d</i> data from 3 to show fluctuations, 0d / 571 or 10d / 615 or 20d / 455 or 30d / 550 ; 3 <i>conclusion 0d / all d with all dR</i> increased (mean) activity when allowed to ripen / dR (bigger than d) or <i>conclusion all dR compared to 0d</i> all, dR data / storage times, with ripening have greater (mean) activity than, 0d / ora ; 4 <i>data comparison d with dR</i> data to compare 10d with 10dR or 20d with 20dR or 30d with 30dR e.g. 20d 455 v 20dR 822 or data to compare 0d / 571 with 10dR / 642 and 20dR / 822 and 30dR / 736 5 error bars overlap (so may be) no <u>significant difference</u> (in catalase activity) between, all / both, groups / in Figure 1.4 / AW ; <i>both = all d data and dR</i>	4	ignore units if seen ignore <i>ref. to dR</i> <table><tr><td>0d 571</td><td>10d and 20d 615 and 455</td></tr><tr><td>0d 571</td><td>10d and 30d 615 and 550</td></tr><tr><td>0d 571</td><td>20d and 30d 455 and 550</td></tr></table> allow dR as AW for ‘time to ripening’ allow stored at 7 degrees / AW for d allow stored at (7 and) 20 degrees / AW for dR 4 ignore means of all d and all dR ; A calculated difference between means allow (95%) confidence intervals / CI / SE / standard error as AW for error bars ignore the size of error bars / SE / CI ignore optimum and highest mean activity A both groups for all groups R results ignore values	0d 571	10d and 20d 615 and 455	0d 571	10d and 30d 615 and 550	0d 571	20d and 30d 455 and 550
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Question	Answer	Marks	Guidance
1(c)(vi)	any two from: (data) not normally distributed ; standard deviations are different ; more than two means (to compare) ;	2	R data is, not continuous / discrete / discontinuous / categoric / qualitative ignore data is, continuous / quantitative A comparing 7 means ignore fewer than 30 values